

GENERAL SPECIFICATION G3.30

Road Work

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G3.30 ROAD WORK

G3.30.1 Preparation of Formation to Receive Road

- (a) The preparation of formation shall be carried out only after completion of all sub grade drainage, piped drains, services and ducts or any other drainage work that might affect the works, unless otherwise agreed by the S.O., and such preparation shall be carried out immediately prior to the laying of the sub base or the road base where no sub base is required.
- (b) The surface of the formation shall be substantially uniform in density throughout its entire width and shall conform to the density requirements for compaction. The surface of subgrade whether on fill or in cut ground shall be compacted to the requirements for earth subgrade as specified in the general specifications. The surface shall be trimmed to conform to the lines, camber, grades and typical cross sections shown on the Drawings, or as directed by the S.O.. The surface shall be constructed to sufficient accuracy to permit the construction of subsequent layers of material to the thickness, cross section surface tolerance and compaction specified but in no case, shall a deviation of more than $\pm 25\text{mm}$ from the levels shown on the Drawings be permitted. Where ruts or other objectionable irregularities occur as a result of construction traffic or other causes, the Contractor shall reshape and recompact the formation to the required density before the sub base or base is placed.

G3.30.2 Earth Subgrade

- (a) Where soft spots and unsuitable materials are present in the sub-grade, they shall be removed and backfilled with approved suitable material and compacted in accordance with the Specification for "Placing of Fill - General". The material used for subgrade shall have a soaked California Bearing Ratio (CBR) value, after the specified compaction, as indicated on the Drawings or if not indicated, not less than 5%. The formation shall then be checked and the final trimmed surface shall be rolled by one pass of a smooth wheeled roller of not less than 5 tonnes or an equivalent vibratory roller prior to the laying of the sub base or base course.
- (b) Notwithstanding the above the top 300mm of any finished fill shall be compacted to a dry density equal to at least 95% of the maximum dry density as determined in the BS Compaction Test as described in Clause 3.3 of BS 1377: Part 4.
- (c) The formation shall be so constructed to drain surface water to the side ditches or other drainage system. If the Contractor allows the moisture content of the completed formation to reach a value above the permitted maximum for the compacted material, the Contractor shall allow the materials to revert to an acceptable moisture content and, if directed by the S.O., make good at his own expense the surface by recompaction before laying the sub base or base.

G3.30.3 Subgrade of Rock

- (a) Where the surface of an excavation in rock has deteriorated and is no longer acceptable as subgrade, the Contractor shall remove all loose rock and replace it with suitable approved material.

G3.30.4 Pavement Thickness

- (a) The thickness of the various layers of the pavement shall be as shown on the Drawings. However, before the construction of the pavement the S.O. may carry out or direct the Contractor to carry out California Bearing Ratio (CBR), soaked or unsoaked tests as described in BS 1377 on the prepared subgrade to determine the strength of the subgrade. All costs for the CBR tests shall be included in the Contract Rates.
- (b) The S.O. may order variations in the thickness of the various layers of the pavement as shown in the typical cross section Drawings as a result of the above tests. Any such changes shall be treated as a variation to the Contract.

G3.30.5 Sub Base Materials

- (a) The sub base material shall be hard, durable and clean. It shall be granular material such as laterite or gravel or crushed rock or sand and shall be free from organic material, clay balls and any other deleterious substances.
- (b) The sub base material shall have a CBR value of not less than 30 when compacted to 95% of the maximum dry density as determined by BS 1377.
- (c) Laterite or crushed rock or gravel shall conform to either of the following gradings:

BS Sieve Size	Percentage by Weight Passing	
	(A)	(B)
40mm	100	-
25mm	75-100	70-100
20mm	60-90	58-75
10mm	45-75	35-65
5mm	30-60	35-65
No. 7 (2.4mm)	20-50	25-50
No. 36 (425 µm)	10-30	15-30
No.200 (75 µm)	0-2	0-2

- (d) The portion passing the No. 36 (425 µm) sieve shall, if it is plastic, have a liquid limit not greater than 35 and a plasticity index not greater than 11.
- (e) Sand shall conform to the gradings given below:

BS Sieve Size	Percentage by Weight Passing
10mm	100
5mm	95-100
No.14 (1.4 mm)	45-80
No.52 (0.3 mm)	10-30
No.100 (150 µm)	2-10

G3.30.6 Alternative Sub Base

- (a) Alternative suitable material may be accepted subject to the prior approval of the S.O.. The proposed sub base material shall have a laboratory controlled unsoaked CBR value not less than 30 when compacted to 95% of maximum dry density as determined by the BS 1377 or the modified AASHO compaction test. The material shall be free from vegetable matter, soft particles or clay lumps and shall conform to one of the following gradings:

BS Sieve Size	Percentage by Weight Passing	
	(A)	(B)
40mm	100	-
20mm	80-100	100
10mm	55-90	80-100
5mm	40-70	55-90
No. 7	12-55	40-70
No. 36	12-30	20-40
No.200	5-20	8-25

- (b) The portion passing the No. 36 sieve shall, if it is plastic, have a liquid limit not greater than 35 and a plasticity index not greater than 11.

G3.30.7 Compaction of Sub Base

- (a) Prior to laying of sub base material, the sub-grade shall be shaped and compacted to the approval of the S.O. and complete for at least 50 m ahead of placing of sub base material.
- (b) The sub base material shall be laid in layers not exceeding 225 mm in uncompacted thickness. At the time of laying and compaction of sub base, the formation shall be dry. Compaction of sub base shall be carried out by adequate compaction equipment approved by the S.O..
- (c) Rolling operations shall begin from the outer edge towards the centre gradually in a longitudinal direction except on super elevated curves where rolling shall begin at the low side and progress towards the high side. Sub base sandy material shall be compacted by use of vibrating equipment.
- (d) Each layer shall be compacted to the requirement specified in Table G3.30-1 and to a compacted density of at least 95% of the maximum dry density as determined by BS 1377.
- (e) If necessary the Contractor shall add water to the sub base material to achieve the required moisture content.
- (f) Sub base material containing excess moisture shall be dried by harrowing, scarifying, ripping or mixing prior to or during compaction. The Contractor shall ensure that the subgrade is not disturbed by the operations carried out to adjust the moisture content of the sub base material.
- (g) The finished sub base shall not vary by more than +/- 20 mm from the grade or level shown on the Drawings. The thickness of the finished sub base shall be on average not less than the required thickness.

G3.30.8 Road Base Materials

- (a) The crushed rock road base to be placed on top of the sub base shall consist of sound durable aggregate material free from clay lumps, organic matter, objectionable coatings or other foreign matter. It shall be laid to give a total compacted thickness, width, grade and levels as shown on the Drawings or as directed by the S.O.. It shall be from a quarry approved by the S.O..
- (b) The road-base material shall be crushed rock of hard durable particles or rock crushed to the correct size, well graded and lie within the following grading limits:

BS Sieve Size	Percentage by Weight Passing
50mm	100
40mm	95-100
20mm	60-80
10mm	40-60
5mm	25-40
No. 7 (2.4mm)	15-30
No. 25 (600 µm)	8-22
No.200 (75 µm)	0-8

- (c) The particle size shall be determined in accordance with the requirements of BS 1377.
- (d) The material shall have a CBR value of not less than 80 when compacted to 98% of the maximum dry density as determined by BS 1377. The flakiness index of the material shall be less than 35%. The water absorption shall be less than 2% and the specific gravity shall exceed 2.40. The Los Angeles Abrasion value of the material shall not exceed 40 or as approved by the S.O..

G3.30.9 Laying and Compaction of Road-Base

- (a) All work on the portion of the subgrade or sub base on which the road-base is to be laid shall be properly shaped and compacted in accordance with the requirements of the Specification before the placing of the road-base material on that portion. The preparation of the subgrade or sub base shall be completed at least 50 m ahead of the placing of the road-base material. The road-base shall be constructed to the full compacted thickness as shown on the Drawings, or as directed by the S.O..
- (b) All material shall be placed and spread evenly to the full width of the road-base. The maximum compacted thickness of any one layer shall not exceed 150mm.
- (c) When the specified compacted thickness of road-base exceeds 150mm, the base shall be constructed in two or more layers of approximately equal thickness. Each layer shall be compacted to the required density and blinded with blinding material as specified before the next layer is placed. Care shall be taken to prevent segregation of the material into the coarse and fine fractions. Segregated surface areas of road-base material contaminated to such an extent that it no longer complies with the Specification shall be removed and replaced with well graded and satisfactory material at the expense of the Contractor.
- (d) The material shall be spread by a motor grader or any other approved mechanical plant.

- (e) Segregation between the coarse and fine particles which results in hungry patches during compaction caused by tipping of road-base material and laterally spreading by dozer or motor grader is to be avoided. Should this occur, all such areas shall be scarified and additional fines added before re-compaction, at the Contractor's own expense.
- (f) Where road-base is spread adjacent to concrete kerbs or gutter sections, extreme care shall be exercised so as not to damage the kerbs or gutters. Any damaged kerbs or gutters shall be removed and replaced at the Contractor's own expense.
- (g) Compaction shall be completed as soon as possible after each layer has been spread and shaped satisfactorily. Each layer shall be thoroughly compacted with suitable compaction equipment to the requirement specified in Table G3.30-1 and to a compacted density of at least 98% of the maximum dry density as determined in accordance with BS 1377.
- (h) Rolling operations shall begin along the edges and progress towards the centre gradually in a longitudinal direction. On super elevated curves, the rolling shall begin at the low side and progress towards the high side. The rolling operation for each layer shall continue until all rolling marks, ridges or cracks are eliminated. Any irregularities, segregation or loose material which may develop in the surface during or after construction shall be corrected or removed and the defective areas made good to the full thickness of the layer and recompacted.
- (i) Special care shall be taken to obtain full compaction in the vicinity of both longitudinal and transverse joints. On completion of compaction the surface of any layer of material shall be well knitted, free from movement under compaction plant and from compaction planes, ridges, cracks or loose material. All loose segregated or otherwise defective areas shall be made good by being removed and replaced with properly graded material.
- (j) Before the laying of each layer of road-base material a partial width of the shoulder, not less than 300mm wide, shall be constructed to the top of each uncompacted layer of the base to act as a haunch to support the edges of the road base during rolling.
- (k) The road-base material shall be maintained at the correct moisture content as necessary for proper compaction. Moisture content control of the road-base material shall be accomplished by sprinkling with water or drying as required, at the Contractor's own expense.
- (l) The surface of the road-base shall be well drained at all times.

G3.30.10 Finished Road-Base Surface

- (a) The finished road-base surface shall be even, free from hungry patches, irregularities, loose material or fines and true to cross section, lines and level. Any irregularities on the finished road base surface of more than 12mm when measured with a straight edge 3 metres long laid either parallel to the centre line of the road or laid transversely, shall be corrected by loosening, adding or removing material, reshaping and recompacting. The finished surface level of the road-base shall not vary from the level shown in the Drawing by more than 12 mm.

G3.30.11 Bituminous Material

- (a) Petroleum Bitumen shall be intended when material is referred to as "Asphalt Cement" "Straight run Bitumen", "Penetration grade bitumen", or by its penetration value. Petroleum Bitumen shall comply with the requirements (for the appropriate grade) given in BS EN 12592.

G3.30.12 Cutback Bitumen

- (a) Cutback bitumen shall be of the rapid curing type or the medium curing type and shall comply with the requirements (for the appropriate grade of cut back bitumen) given in BS EN 12592.

G3.30.13 Bitumen Emulsion

- (a) Bitumen Emulsion shall be of the anionic and or cationic type and shall comply with the requirements (for the appropriate grade) given in BS 434.

G3.30.14 Storage, Handling and Heating of Bitumen

- (a) Bituminous materials shall be handled and stored with due regard for safety and in such a way that at the time of use in the work the material conforms to the Specification. In particular, bitumen emulsion shall be handled with care and not subjected to mechanical shocks or extremes of temperature likely to cause separation of bitumen. Bitumen emulsion showing signs of separation shall not be used.
- (b) Bitumen shall be heated in an approved type of bitumen heater having a capacity commensurate with the output required. The heater shall be fitted with a thermometer accurately registering the temperature of the bitumen and so situated as to be readily accessible.
- (c) Great care must be ensured that the bitumen be heated to a temperature within the range specified for its grade. Overheated bitumen shall, if the S.O. so direct, be disposed of at the Contractor's expense.

G3.30.15 Depot Tray Test

- (a) The Contractor shall provide a set of 50mm wide troughs as described in BS 1707 and shall carry out Depot Tray Test A in the presence of the S.O. of the bitumen distributor in accordance with BS 1707.
- (b) This test shall be carried out prior to using the distributor for the first time on the Contract and at such intervals thereafter as may be required by the S.O..
- (c) Should the quantity of bitumen in any trough or troughs be found to lie outside the prescribed tolerance from the average, the Contractor shall carry out such adjustments to the distributor, or, including the replacement of any faulty jets, as may be necessary to achieve compliance with the test requirements.

G3.30.16 Bitumen Prime Coat

- (a) The prime coat shall be a medium curing cut-back bitumen of grade M.C.O. or M.C.I. The actual grade to be used shall determine the penetration of each grade.

G3.30.17 Blinding

- (a) Material for blinding shall be clean, sharp and free from organic matter and with not more than 45% of material passing No.25 BS Sieve. Not more than 5% of material used for blinding shall be retained on the 6.35mm BS Sieve.
- (b) Alternatively, grit arising from the crushing of rock may be used, subject to the approval of the S.O. regarding its grading.

G3.30.18 Surface Preparation for Prime Coat

- (a) Immediately before the application of the prime coat the surface of the road base shall be brushed clean and free from all dust, dirt, dried or wet mud, organic matter or any other loose material to the satisfaction of the S.O..
- (b) If considered necessary by the S.O., the surface of the road base shall be slightly damped before the application of the prime coat. Moisture control of the road base shall be carried out by the Contractor using a wetting agent and method approved by the S.O. and to a level of dampness to the satisfaction of the S.O..
- (c) Preparation of the surface as above shall not precede the application of bitumen prime coat by more than 450 m. Should any part or parts of the prepared surface become contaminated or unsuitable for the application of the prime coat, due to lack of care on the Contractor's part to protect the surface or due to prolonged exposure, the surface shall be restored or re-prepared to the satisfaction of the S.O..

G3.30.19 Application of Prime Coat

- (a) The prime coat shall be applied uniformly over the full width of the prepared road base by means of an approved pressure load mobile distributor. The pressure feed mobile distributor shall be fitted with a spray bar through which the hot bitumen can be circulated while the distributor is not spraying. The jets shall be so arranged in the spray bar that they are effectively preheated to the same temperature as the circulating bitumen.
- (b) The bitumen shall be applied at a temperature in accordance with the table below and at the rate of spray as directed by or agreed with the S.O. after carrying out Site tests, but not to be less than 440m L/m².

Grade Of Bitumen	Temperature Of Application °C
M.C.O.	40 - 60
M.C.I.	60 - 70

G3.30.20 Spraying Procedure of Prime Coat

- (a) Immediately before any spraying run commences the spray bar shall be tested by allowing all the jets to discharge simultaneously for about 30 seconds with a slotted metal tray placed beneath the spray bar whilst a visual inspection is made. Any jet which is seen not delivering a uniform spray cone shall be cleaned or replaced. After testing the spray bar the distributor shall proceed with the spraying run without delay.

- (b) Any areas insufficiently covered shall be resprayed by spray lance to the satisfaction of the S.O.. No spraying of bitumen shall be carried out during rain; or when rain is imminent, or immediately after rain until the road base has dried.
- (c) After spraying the surface shall be left for a minimum period of three (3) hours before blinding. Blinding shall be applied so as to obtain a uniform covering without evidence of bare or fat patches of bitumen. The blinding shall be rolled with not more than two passes of an approved roller.

G3.30.21 Protection of Prime Coat

- (a) After laying the prime coat, barriers shall be erected to keep the area surfaced free from traffic for such period as the S.O. may direct.

G3.30.22 Dense Bitumen Macadam Surfacing

- (a) Aggregate, Binder and Composition
 - (i) The binder course and wearing course shall each have a thickness as shown on the Drawings and be laid in accordance with BS EN 12591 after construction of road base and bitumen prime coat or tack coat.
- (b) Coarse Aggregate
 - (i) The coarse aggregate shall be natural aggregate substantially free from material passing a BS 3.35mm Sieve. It shall be hard, clean, durable crushed rock of the granite group or of any other rock group as may be approved by the S.O.. Only granite shall be used for the wearing course.
- (c) Fine Aggregate
 - (i) The fine aggregate shall consist of crushed rock or clean sand, substantially all of which shall pass a 3.35mm BS Sieve. If sand is used, the content of silt, loam and clay shall not exceed 3% by weight of the fine aggregate;
 - (ii) If added filler is used in the bitumen macadam, it shall consist of crushed rock, hydrated lime, portland cement, or other material approved by the S.O.. At least 75% of it shall pass a 75 μ m BS Sieve.
- (d) Binder
 - (i) The binder shall be straight run bitumen, which shall have a penetration at 25oC between 80 and 100.
- (e) Composition of Mixtures
 - (i) The composition of freshly mixed material for the bitumen macadam shall comply on analysis with the following requirements:

1) Binder Course

Passing BS Sieve	38mm nominal size aggregate % by weight
63.0mm	-
50.0mm	100
37.5mm	90-100
28.0mm	50-80

Passing BS Sieve	38mm nominal size aggregate % by weight
14.0mm	10-30
3.35mm	0-10
Binder content as found by analysis	3.0-3.8

2) Wearing Course

Passing BS Sieve	38mm nominal size aggregate % by weight
28mm	100
20mm	95-100
14mm	70-90
10mm	55-75
6.3mm	40-60
3.35mm	25-40
1.18mm	14-30
75µm	3-6
Binder content as found by analysis	4.4-5.4

- (ii) When material other than limestone is the aggregate, the material passing 75m BS test sieve shall include portland cement or hydrated lime to the amount of 2% by mass of the total aggregate.
- (iii) The exact bitumen content depending on density, shape and grading of the aggregate will be determined by the S.O..

G3.30.23 Mixing of Bitumen Macadam

- (a) The aggregate shall be surface dried, and the aggregate and binder separately heated to the following temperatures before entering the mixer.

Aggregate	120-148 °C
Binder	130-162 °C

- (b) It is particularly important to avoid excessive heating of the binder, and excessively high temperatures of the aggregate at the time of mixing, as these will adversely affect the quality of the resultant macadam.
- (c) The materials, including any added filler, shall be weighed or measured in a mechanical mixer and thoroughly mixed in such a manner that all particles of the aggregates are completely and uniformly coated.
- (d) The mixing plant shall have a capacity sufficient to supply the paver on the road continuously when spreading the mix at normal speed and required thickness.

G3.30.24 Transport of Bitumen Macadam

- (a) The bitumen macadam shall be transported from the manufacturing plant to the Site in clean vehicles and shall be protected against adverse weather conditions by means of tarpaulins. The use of coated dust, oil or water on the interior of vehicles to facilitate discharge of the bitumen macadam may be permitted, but the amount shall be kept to a minimum and any excess shall be removed by tipping or brushing.
- (b) It is particularly important that bitumen macadam which is to be laid warm shall be protected to minimise loss of heat during transit so that all material is delivered in a condition suitable for spreading and compacting.

G3.30.25 Sampling and Testing

- (a) Sampling and testing of the aggregate and mixed materials shall be carried out in accordance with the relevant LTA standards.

G3.30.26 Bitumen Macadam Layer Thickness and Tolerance

- (a) The binder course and wearing course shall be laid by machine in one single course each and the compacted thickness of the individual courses shall be as shown on the Drawings.
- (b) The accuracy of finish in the longitudinal direction shall be determined by measuring the gap under a 3m straight edge, placed in any position on the road surface parallel to the centre line.
- (c) The gap at any place between the points at which the straightedge is in contact with the road shall not exceed the following limits:

Binder course :	10mm
Wearing course :	16mm

- (d) The transverse profile shall conform to a similar standard of accuracy, using a correctly shaped template instead of a straightedge.
- (e) The top surface of the binder and wearing courses shall be finished to the specified levels, grades and cross falls within a tolerance of $\pm 6\text{mm}$. The finished surface of the wearing course shall in addition be finished flush with kerbing and with a constant fall so that no ponding will occur.

G3.30.27 Laying of Bitumen Macadam

- (a) The bitumen macadam shall be laid by machine. Works shall not proceed under unsuitable weather conditions. The machine shall be of a type preferably self-propelled and capable of laying bitumen macadam continuously so as to produce automatically an even, smooth and compact surface to the required widths, thicknesses and cross falls without segregation of the bitumen macadam.
- (b) A fully trained and experienced operator shall be in direct charge of the machine.
- (c) The bitumen macadam shall be supplied continuously to the machine. Material supplied to the machine shall be laid as soon as possible after delivery.

- (d) The machine shall be used only when it is in a serviceable condition. Bituminous material remaining in the hoppers, conveying and spreading mechanisms, tampers and screeds, shall be cleaned off at the end of each working day. On no account shall cleaning solvent be allowed to come into contact with any road surfacing material. Accumulations of material not of the specified quality shall be removed from the Site. The machine shall be operated to avoid, as far as possible, dragging of the bitumen macadam.
- (e) On carriageways, narrow strips remaining alongside machine work shall be as far as possible hand laid and rolled at the same time as the machine laid work and allowance shall be made for extra compaction of hand laid strips.
- (f) Continuous inspection of the finished surface as it is laid shall be carried out and any defects immediately rectified before any rolling takes place.

G3.30.28 Bitumen Tack Coat

- (a) The surface of the dense bitumen macadam binder course shall be cleaned immediately prior to the application of the tack coat. The bitumen tack coat material shall be applied by means of distributor at the rates directed by the S.O., but not to be less than 270 mL/m² and at temperatures within the range required.

G3.30.29 Joints

- (a) When a transverse construction joint is to be made, the material shall be cut back to a vertical face of the full depth of the course being laid and painted with bitumen or bitumen emulsion as specified before any new material is laid.
- (b) On longitudinal joints a complete bond of the two strips after compaction shall be ensured. If this is not possible, the edge shall be cut back and painted as above.

G3.30.30 Laying Around Manhole Covers

- (a) When laying around manhole covers and similar fittings, those parts against which the bitumen macadam is to abut shall be cleaned and painted with bitumen or bitumen emulsion as specified prior to laying operations. The bitumen macadam shall be tamped so that after final compaction, the finished surface is level with, or slightly proud of, such fittings.

G3.30.31 Compaction

- (a) As soon as rolling can be effected without causing undue displacement, the bitumen macadam shall be uniformly compacted by rolling to a density of not less than 98 % of the Marshall Density of the mix. Compaction shall be carried out with a roller or rollers of weight not less than 6 tonnes nor exceeding 10 tonnes, and the width of a roller shall not be less than 1.0m. The bitumen macadam shall be rolled in a longitudinal direction, working from the sides to the centre of the carriageway, overlapping on successive passes by at least one half of the width of the rear roll. The roller shall be fitted with a quick reverse and smooth acting clutch. If directed by the S.O. suitable pneumatic tyre rollers shall be provided by the Contractor at no extra cost.
- (b) Rollers shall not be allowed to stand on newly laid macadam as there is a risk that the surface will be deformed.

- (c) Traffic shall not be allowed to pass over the surface until the bitumen macadam has been completed and is adequately set.

G3.30.32 Road Shoulders and Stabilised Access

- (a) Road shoulders and stabilised access shall be constructed to the lines, grades, levels, dimensions and cross sections shown on the Drawings and as directed by the S.O..
- (b) The shoulder shall be constructed on a previously prepared subgrade finished as required in the Specification. The work shall be carried out in conformity with the requirements for simultaneous construction of sub base or shoulder base where required and as approved by the S.O.. The material for shoulder sub-base construction shall be of well graded laterite, gravel mining ballast or other suitable local material approved by the S.O. and conform to the requirements of road sub base material and the following additional requirements.
- (i) For material passing through No 36 sieve (425µm)
- | | | |
|---|---|----|
| 1) Minimum Plasticity Index | : | 10 |
| 2) Maximum Plasticity Index | : | 30 |
| 3) Minimum CBR at 95% maximum dry density | : | 20 |
- (c) Shoulder sub-base shall be placed and compacted in layers not exceeding 200mm thickness before compaction. The layers shall be compacted to 95% of maximum dry density as determined in accordance with BS 1377.

G3.30.33 Shoulder Base Selected Fill Grading

- (a) Selected fill material is to be used as shoulder base material over the compacted sub-base at the edge of the pavement and the shoulder and any other areas shown on the Drawings or as directed by the S.O..
- (b) It shall comprise clean, hard, durable, well graded sandy gravels obtained from crushed rock and shall be free from any organic matter or deleterious material.
- (c) The grading of the material shall fall within the following limits:

BS Sieve	% Passing by weight
50mm	100
40mm	80-100
20mm	45-70
10mm	25-50
5mm	15-30
2.4mm	12-20
600µm	6-12
75µm	2-8

- (d) Alternative suitably graded material may be accepted by the S.O. or used as directed by the S.O..

G3.30.34 Lateral Shoulder Subsoil Drains

- (a) Shoulder subsoil drains shall be constructed along the road shoulders as shown on the Drawings or as directed by the S.O..
- (b) Such lateral shoulder subsoil drains shall be constructed of graded selected fill to provide free drainage outlet from the pavement sub base and shall be constructed to the dimensions and material as shown on the Drawing or as directed by the S.O..

G3.30.35 Refilling of Excavated Pipe Trenches for Roadways

- (a) In roads and in all other places where the S.O. considers mechanical compaction is required, the suitable material above the pipe shall be spread in layers of not greater than 150 mm and thoroughly consolidated by mechanical means as listed out in Table G3.30-1.
- (b) As soon as the compaction of sub base is completed the road base will be laid and compacted, before proceeding with the laying of the binder and wearing courses. Any depressions found shall be made up with material of the same kind as that of the original surface all at the Contractor's own expense.

G3.30.36 Joining to Existing Roads

- (a) Where the new carriageway abuts and joins on to existing roads and where directed by the S.O. the existing roads shall be scarified, adjusted and reshaped to roads conform to existing and new cambers or cross-falls.

G3.30.37 Notes Relating to Table G3.30-1

- (a) The notes given in this clause and Table G3.30-1 are for guidance only and shall not relieve the Contractor from his basic obligation to produce work to the required standard of the Specification.
 - (i) The depth of a compacted layer is the height by which an embankment is raised by each successive compacted layer.
 - (ii) Number of passes is the number of times that each point on the surface of the layer being compacted has been traversed by the item of compaction plant. In assessing the number of passes of pneumatic tyred rollers to meet the requirements of Table G3.30-1 the effective width shall be the sum of the widths of the individual wheel tracks together with the sum of spacing between the wheel tracks provided each spacing does not exceed 288mm for Table G3.30-1. When the spacing exceeds the above, the effective width shall be the sum of the widths of the individual wheel tracks.
 - (iii) The force per 100mm width is the total force (weight) on the roll divided by the total roll width. Where a smooth wheeled roller has more than one axle the machine will be assessed on the basis of the axle giving the highest value of force per 100mm width.
 - (iv) For pneumatic tired rollers, wheel load is the total weight of the roller divided by the number of wheels.
 - (v) Vibratory rollers are self-propelled or towed having means of applying mechanical vibration to one or more rolls.

- 1) The requirement for vibratory rollers are based on the use of the lowest gear on a self-propelled machine and towing speed of 1.5 - 2.5 km per hour for a towed machine. If higher gears or speeds are used an increased number of passes shall be provided in proportion to the increase in speed of travel.
 - 2) Vibratory rollers operating without their vibration mechanism in use will be classified as smooth wheeled rollers.
 - 3) Vibratory rollers shall be operated with their vibration mechanism operating only at the frequency of vibration recommended by the manufacturers. All such rollers shall be equipped with a device automatically indicating the frequency at which the mechanism is operating.
 - 4) Combination of different types of plant will be permitted in which case the number of passes for each type shall follow such proportion of the appropriate number as will together produce the same total compactive effort as any type operated singly.
- (vi) Vibrating plate compactors are machines having a base plate to which is attached a source of vibration consisting of one or two eccentrically weighted shafts:
- 1) The static pressure under the plate of a vibrating plate compactor is calculated by dividing the total weight of the machine in working order by the area in contact with compacted soil.
 - 2) Vibrating plate compactors shall be operated at the frequency of vibration recommended by the manufacturers. They shall normally be operated at travelling speeds of less than 15 m/min but if higher speeds are necessary the number of passes shall be increased in proportion to the increase in speed of travel.
- (vii) Vibro tampers are machines in which an engine driven reciprocating mechanism acts on a spring system, through which oscillations are set up in a base plate.
- (viii) Power rammers are machines which are actuated by explosions in an internal combustion cylinder, each explosion being controlled manually by the operator.
- (ix) In the case of power rammers one pass will be considered as made when the compacting shoe has made one strike on the area in question.
- (x) Where combinations of different types or categories of plant are used, the compaction requirements shall be:
- 1) The depth of layer shall be that for the type of plant requiring the least depth of layer;
 - 2) The number of passes shall be that for the type of plant requiring the greatest number of passes; and
 - 3) However, where the Contractor uses a lighter type of plant to provide some preliminary compaction only these requirements to assess the effective compaction could be disregarded.

Table G3.30-1: COMPACTION PROCEDURE FOR UNBOUND SUB-BASE AND ROAD-BASE MATERIALS
FOR GUIDANCE ONLY

Type of compaction plant	Category	Minimum number of passes		
		< 110mm layer	< 150mm layer	< 225mm layer
Smooth-wheeled roller	Force per 100mm width (kN)			
	2.6-5.2 > 5.2	16 8	Unsuitable 16	Unsuitable Unsuitable
Pneumatic-tyred roller	Wheel load (t)			
	4-6	12	Unsuitable	Unsuitable
	6-8	12	Unsuitable	Unsuitable
	8-12	10	16	Unsuitable
	> 12	8	12	Unsuitable
Vibratory roller	Static force per 100mm width (kN)			
	0.71-1.25	16	Unsuitable	Unsuitable
	1.26-1.75	6	16	Unsuitable
	1.76-2.30	4	6	10
	2.31-2.80	3	5	9
	2.81-3.50	3	5	8
	3.51-4.20	3	4	7
	4.21-4.90	2	4	6
Vibrating-plate compactor	Static force under base plate (kN/m ²)			
	13.8-17.2	8	Unsuitable	Unsuitable
	17.2-20.7 > 20.7	5 3	8 6	Unsuitable 10
Vibro-tamper	Mass (kg)			
	50-60	4	8	Unsuitable
	65-75	3	6	10
	> 75	2	4	8
Power rammer on dropping weight compactor	Mass (kg)			
	100-500 > 500	5 5	8 8	Unsuitable 12

END OF SECTION